

Immunogenicity and safety of a booster dose of the 13-valent pneumococcal conjugate vaccine in children primed with the 10-valent or 13-valent pneumococcal conjugate vaccine in the Czech Republic and Slovakia[☆]



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ARTICLE INFO

Article history:

Received 24 April 2017

Received in revised form 25 July 2017

Accepted 27 July 2017

Available online 7 August 2017

Keywords:

Pneumococcal conjugate vaccine

Interchangeability

Immunisation schedule

Immunogenicity

Serotype 19A

ABSTRACT

Background: Although both the 13-valent pneumococcal conjugate vaccine (PCV13) and the 10-valent pneumococcal non-typeable *Haemophilus influenzae* protein D-conjugate vaccine (PHiD-CV) are widely used, it is unclear how interchangeable they are in terms of immunogenicity.

Methods: Two phase 3, open-label, multicenter studies were conducted to assess the immunogenicity and safety of a booster dose of PCV13 in children primed with PHiD-CV or PCV13. In the Czech Republic, 12–15-month-old children received a PCV13 booster after 3-dose priming with either PHiD-CV or PCV13. In Slovakia, 11–12-month-old children received PCV13 following 2-dose priming with either PHiD-CV or PCV13. Serum IgG concentrations were assessed by enzyme-linked immunosorbent assay and functional antibodies were assessed by opsonophagocytic assay (OPA) before the booster and at 1 and 12 months afterward. The primary objective of these studies was to assess non-inferiority of OPA titers for serotype 19A in PHiD-CV-primed subjects compared to those in PCV13-primed children 1 month post-booster.

Results: A total of 98 subjects in the Czech Republic and 89 subjects in Slovakia were included. One month after the PCV13 booster dose, the IgG and OPA immune responses to serotype 19A in subjects primed with 2 or 3 doses of PHiD-CV were non-inferior to those in subjects primed with PCV13. Non-inferior and persistent immune responses to most other vaccine serotypes were also observed after the PCV13 booster in PHiD-CV-primed subjects. No safety issues were raised in either study.

Conclusions: Overall, robust IgG and OPA immunological responses were observed after booster vaccination with PCV13 in children primed with 2 or 3 doses of PHiD-CV or PCV13, including for serotypes not included in PHiD-CV. These results suggest that these vaccines are interchangeable in terms of safety and immunogenicity and that PCV13 can be used as a booster in the context of mixed schedules. (EudraCT numbers: 2012-005366-35 and 2012-005367-27).

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Abbreviations: AE, adverse event; CI, confidence interval; ELISA, enzyme-linked immunosorbent assay; GMC, geometric mean concentration; GMT, geometric mean titer; PCV7, 7-valent pneumococcal conjugate vaccine; PCV13, 13-valent pneumococcal conjugate vaccine; PHiD-CV, pneumococcal non-typeable *Haemophilus influenzae* protein D-conjugate vaccine; SAE, serious AE.

[☆] Presented in part at the Annual Meetings of the European Society for Paediatric Infectious Diseases in Leipzig, Germany, in May 2015 and in Brighton, the United Kingdom, in May 2016; and at the International Symposium on Pneumococci and Pneumococcal Diseases in Glasgow, the United Kingdom, in June 2016.

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1. Introduction

Since the introduction of the first pneumococcal conjugate vaccine (PCV) in 2000, invasive and non-invasive diseases caused by *Streptococcus pneumoniae* vaccine serotypes have dramatically decreased worldwide [1,2]. This 7-valent vaccine (PCV7; Pfizer Inc.) included the capsular polysaccharides of the seven most frequent pneumococcal serotypes in the United States at the time of licensure (serotypes 4, 6B, 9V, 14, 18C, 19F, and 23F) [3]. However, a small but significant rise in infections caused by non-PCV7 serotypes, notably serotype 19A, occurred in several countries, thereby eroding the impact of PCV7 [4–6]. The burden of disease caused by serotype 19A is of particular concern because of its invasive nature and antibiotic resistance profile [7]. To limit the rise of infections by non-vaccine serotypes and increase overall serotype coverage, second-generation PCVs with extended valencies became available in 2009 and have replaced PCV7. The 10-valent pneumococcal non-typeable *Haemophilus influenzae* protein D-conjugate vaccine (PHiD-CV; GSK) includes the PCV7 serotypes plus serotypes 1, 5, and 7F, whereas the 13-valent PCV (PCV13; Pfizer Inc.) includes the PHiD-CV serotypes plus serotypes 3, 6A, and 19A. Both PHiD-CV and PCV13 have been licensed for protection against invasive pneumococcal disease on the basis of immunological non-inferiority compared to the licensed PCV7 [8].

Both vaccines were originally licensed in a 4-dose schedule consisting of 3 primary doses plus a booster dose administered at least 6 months after the third dose (3 + 1 schedule) [9]. Several alternative schedules are also used, without statistically significant differences in antibody concentrations after the booster dose for almost all serotypes [10,11]. One consists of 2 primary doses given 2 months apart, followed by a booster dose at least 6 months after the second dose (2 + 1 schedule) [9]. This schedule was first evaluated and implemented in the United Kingdom [12] and subsequently in several other countries [6]. In developing countries, a 3-dose primary schedule without a booster dose (3 + 0 schedule) is often used to match the schedules of the Expanded Programme on Immunisation [13].

Although immunogenicity of both PHiD-CV and PCV13 has been compared to that of PCV7 [14,15], only a few studies have directly compared PHiD-CV and PCV13 [16–18]. In this study, we compared the immunogenicity and safety of PCV13 when administered as a booster in children primed with PHiD-CV or PCV13, either as a 2- or 3-dose schedule.

2. Materials and methods

2.1. Study design and subjects

These were two phase 3, multicenter, open-label studies conducted in 10 centers in the Czech Republic (EudraCT, 2012-005366-35) and in 8 centers in Slovakia (EudraCT, 2012-005367-27) to assess immunogenicity and safety of alternative vaccination schedules with pneumococcal conjugate vaccines. Both study protocols were approved by the relevant independent ethics committees and the studies were conducted in accordance with the Declaration of Helsinki and Good Clinical Practice guidelines. Written informed consent was obtained from parents or legal representatives of all children before enrolment.

In the Czech Republic (3 + 1 schedule), healthy children were eligible if they were 12–15 months of age and had completed a 3-dose vaccination course before 7 months of age with either the 10-valent PCV (PHiD-CV; *Synflorix*[™], GSK, Rixensart, Belgium) or the 13-valent PCV (PCV13; *Prevenar*/*Prevnar* 13[™], Pfizer, New York, NY). In Slovakia (2 + 1 schedule), healthy children were eligible if they were 11–12 months of age and had completed a 2-dose

vaccination course before 6 months of age with either PHiD-CV or PCV13.

Children were excluded if they had received a previous booster dose against *S. pneumoniae*, any investigational product (drug or vaccine) within 30 days preceding the study vaccination, immunoglobulins or any blood product within 3 months preceding the study vaccination (or were to receive these during the study period), a vaccine (except licensed influenza or diphtheria-tetanus-acellular pertussis vaccines) within 30 days before or after the study vaccination, immunosuppressants or other immunomodifying drugs for >14 days since birth; had any confirmed or suspected immunodeficient condition or family history of immunodeficiency, serious chronic disease, history of neurological disorders or seizures; had any contraindications to vaccination such as allergies; had acute infection or fever (oral, axillary, or tympanic temperature > 37.5 °C or rectal temperature > 38.0 °C) at enrollment.

2.2. Study vaccines

All children received one booster dose of PCV13 by intramuscular injection (0.5 mL) in the anterolateral region of the thigh or the deltoid region. This vaccine contains capsular polysaccharides of serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23 F, all conjugated to the diphtheria CRM₁₉₇ carrier protein. Co-administration of the routine diphtheria-tetanus-acellular pertussis-hepatitis B-inactivated polio-*H. influenzae* type b vaccine (DTaP-HBV-IPV/Hib; *Infanrix hexa*[™]; GSK) was allowed in the Czech Republic and was mandatory in Slovakia.

2.3. Assessment of immunogenicity

Blood samples were taken for immunogenicity analyses before the booster dose (study month 0) and 1 and 12 months after the booster dose. Sera were stored at –20 °C until analysis. Analyses were performed at the World Health Organization pneumococcal serology reference laboratory (University College London, United Kingdom). Immunoglobulin G (IgG) serum concentrations specific for the 13 vaccine serotypes were measured using an enzyme-linked immunosorbent assay (ELISA) after adsorption with cell-wall and 22F polysaccharides to increase the assay specificity [19]. A standardized opsonophagocytic assay (OPA) was used to measure functional antibodies against the same serotypes [20]. The OPA titer was defined as the reciprocal of the lowest serum dilution that induces ≥50% bacterial cell death compared to the assay control. The OPA response to vaccine-related serotype 6C [21] was also measured, although not planned in the protocol.

2.4. Assessment of safety and reactogenicity

Solicited local reactions (pain, redness, and swelling at the injection site) and solicited systemic reactions (drowsiness, fever, irritability/fussiness, and loss of appetite) were recorded by parents on diary cards for 4 days after vaccination (days 0–3). Unsolicited adverse events (AEs) were recorded for 31 days (days 0–30) after vaccination and serious adverse events (SAEs) were recorded over the entire study period. The intensity of solicited reactions, and AEs was scored on a scale from 1 (mild) to 3 (severe). Redness and swelling at the injection site were scored as 1 for >0 to ≤20 mm diameter, 2 for >20 to ≤30 mm, and 3 for >30 mm. Fever was defined as rectal temperature ≥38.0 °C or oral, axillary, or tympanic temperature ≥37.5 °C. Grade 3 fever was defined as rectal temperature >40 °C or axillary, oral, or tympanic temperature >39.5 °C. Pain was scored as 1 for minor reaction to touch, 2 for crying on touch, and 3 for crying when limb is moved or spontaneously painful. The other solicited reactions were scored as 1 if

they did not affect normal activity, 2 if they interfered with normal activity, and 3 if they prevented normal daily activity. All solicited local reactions were considered vaccination-related. Causality of solicited systemic reactions, unsolicited AEs, and SAEs was determined by the investigators.

2.5. Statistical analysis

Safety was analyzed in the full analysis set, which included all subjects who received a booster dose of PCV13. Immunogenicity was analyzed in the per protocol set, which included all subjects who received the booster dose of PCV13 according to protocol and who had at least one valid IgG or OPA result.

At each visit and for each serotype, antibody geometric mean concentrations (GMCs) and OPA geometric mean titers (GMTs) were determined with 2-sided 95% confidence intervals (CI). The percentages of participants reaching the predefined immunological thresholds (OPA titer ≥ 8 and IgG concentration $\geq 0.35 \mu\text{g/mL}$) were also calculated [22,23]. No formal statistical tests were performed for safety data.

The primary objective of the studies was to assess non-inferiority of the OPA GMT to serotype 19A in subjects primed with PHiD-CV compared to those primed with PCV13, 1 month after the PCV13 booster dose administered at 12–15 months in the Czech Republic (3 + 1) and at 11–12 months in Slovakia (2 + 1). The secondary confirmatory objective was to assess non-inferiority of the IgG response to serotype 19A 1 month post-booster. Non-inferiority was demonstrated if the lower limit of the 2-sided 95% CI for the OPA GMT ratio or the IgG GMC ratio (PHiD-CV group over PCV13 group) was >0.5 . Other secondary exploratory objectives included IgG and OPA responses against the other serotypes at month 1 and against all serotypes at month 12, and assessment of safety and reactogenicity of the PCV13 booster dose.

In both studies, a planned sample size of 92 subjects (46 participants in both vaccine groups) was calculated to provide 90%

power for assessing non-inferiority in pneumococcal serotype 19A antibody response rate with a limit of 10% at a 2-sided 5% significance level, assuming a 98% response rate in the PHiD-CV group and a 10% drop-out rate. All significance tests were two-tailed. A P -value ≤ 0.05 was considered to indicate statistical significance. Statistical analyses were performed with SASTM version 9.1 or higher (SAS Institute, Cary, NC).

3. Results

3.1. Subjects

The study in the Czech Republic was conducted between March 12, 2013 and October 7, 2014 and the study in Slovakia between March 5, 2013 and November 14, 2014. Overall, 98 children were enrolled in the Czech Republic and 89 in Slovakia (Fig. 1). All subjects in both studies received the PCV13 booster dose and were included in the full analysis set. Twelve months post-booster, 94.1–98.0% of subjects per group were included in the per protocol set. In the Czech Republic, DTaP-HBV-IPV/Hib was co-administered in 44 subjects (83.0%) in the PHiD-CV-primed group and 39 (86.7%) in the PCV13-primed group. In Slovakia, one subject in the PCV13 group did not receive DTaP-HBV-IPV/Hib concomitantly with PCV13, as stipulated in the protocol, and was therefore excluded from the per protocol set.

In both studies, most participants were male and Caucasian (Table 1). In the Czech Republic, 73.3% of the subjects in the PCV13 group were male compared to 56.6% in the PHiD-CV group. Per study design, participants were older in the Czech Republic (mean age, 12.6 months) than in Slovakia (mean age, 11.3 months).

3.2. Immune responses against serotype 19A

Before the booster dose, OPA GMTs against serotype 19A were higher in children primed with PCV13 than those primed with

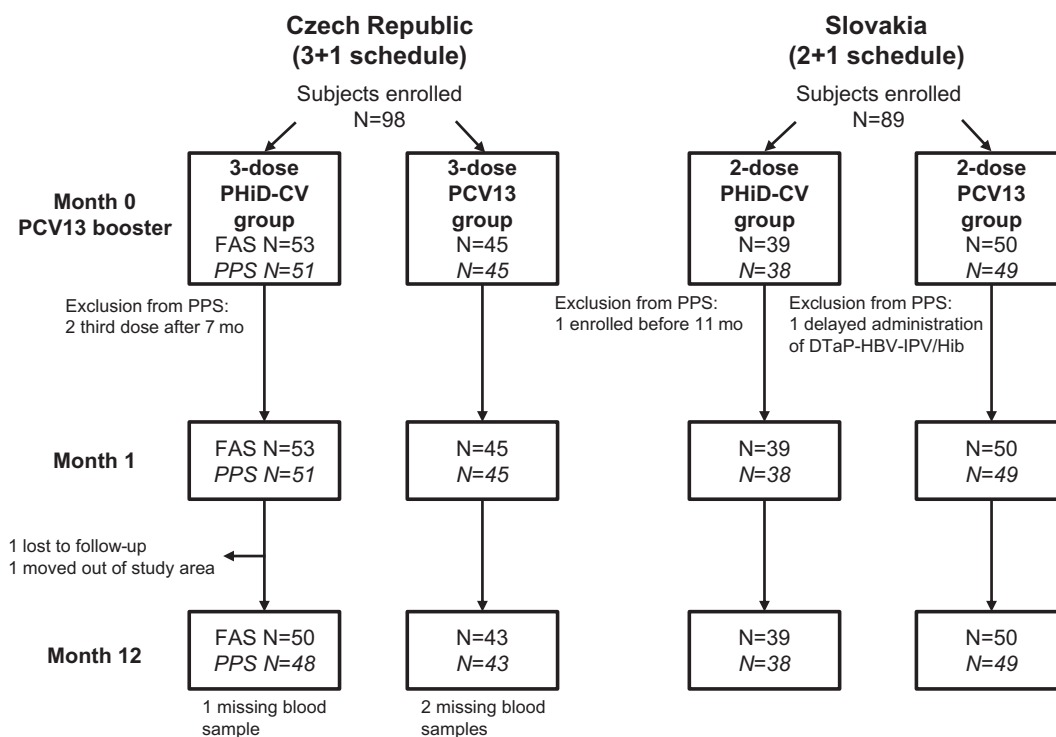


Fig. 1. Flowchart of participants. FAS, full analysis set; DTaP-HBV-IPV/Hib, diphtheria-tetanus-acellular pertussis-hepatitis B-inactivated polio-*Haemophilus influenzae* type b vaccine; PCV13, 13-valent pneumococcal conjugate vaccine; PHiD-CV, 10-valent pneumococcal non-typeable *Haemophilus influenzae* protein D conjugate vaccine; PPS, per protocol set.

Table 1
Baseline demographic characteristics of participants.

Characteristics, n (%)	3 + 1 schedule (Czech Republic)			2 + 1 schedule (Slovakia)		
	PHiD-CV group N = 53	PCV13 group N = 45	Total N = 98	PHiD-CV group N = 39	PCV13 group N = 50	Total N = 89
Age (months)						
Mean (SD)	12.9 (1.1)	12.3 (0.6)	12.6 (1.0)	11.3 (0.6)	11.4 (0.5)	11.3 (0.5)
Range	12.0–16.0	12.0–14.0	12.0–16.0	10.0–13.0	11.0–13.0	10.0–13.0
Sex						
Female	23 (43.4%)	12 (26.7%)	35 (35.7%)	16 (41.0%)	21 (42.0%)	37 (41.6%)
Male	30 (56.6%)	33 (73.3%)	63 (64.3%)	23 (59.0%)	29 (58.0%)	52 (58.4%)
Race						
Caucasian	53 (100.0%)	45 (100.0%)	98 (100.0%)	39 (100.0%)	49 (98.0%)	88 (98.9%)
Other	0	0	0	0	1 (2.0%)	1 (1.1%)

PCV13, 13-valent pneumococcal conjugate vaccine; PHiD-CV, 10-valent pneumococcal non-typeable *Haemophilus influenzae* protein D conjugate vaccine; SD, standard deviation.

PHiD-CV irrespective of schedule (Supplementary Table 1). One month post-booster, OPA GMTs increased markedly irrespective of vaccine and schedule and, subsequently, decreased between months 1 and 12. Post-booster 19A OPA GMTs were comparable in PHiD-CV and PCV13 groups irrespective of the priming schedule. One month after the booster dose, all subjects had 19A OPA GMTs ≥ 8 but, at 12 months, fewer subjects had GMTs ≥ 8 in the PHiD-CV groups (84.3–86.8%) than in the PCV13 groups (93.9–97.8%) (Supplementary Table 2).

One month post-booster, the OPA GMT ratio (PHiD-CV group over PCV13 group) for serotype 19A was 1.24 (95% CI, 0.84–1.82) with the 3 + 1 schedule and 0.81 (95% CI, 0.54–1.20) with the 2 + 1 schedule (Fig. 2). The OPA titers 1 month after the booster in PHiD-CV-primed children met the a priori non-inferiority criteria.

Twelve months post-booster, 19A OPA titers in PHiD-CV-primed children were inferior to those in PCV13-primed children in both studies.

One month post-booster, 19A IgG GMCs were similar between the two schedules, increased 15–29-fold and 12–24-fold with the 3 + 1 and the 2 + 1 schedule, respectively, and decreased between months 1 and 12 (Supplementary Table 3). All subjects at month 1, and between 82.4% and 98.0% at months 12, had GMCs ≥ 0.35 mg/mL (Supplementary Table 4). In both studies, the percentages were comparable between groups. The mean IgG GMC ratio (PHiD-CV group over PCV13 group) for serotype 19A at 1 month post-booster was 1.36 (95% CI, 0.90–2.08) with the 3 + 1 schedule and 1.25 (95% CI, 0.83–1.87) with the 2 + 1 schedule (Fig. 3). Therefore, the antibody concentrations in PHiD-CV-primed children

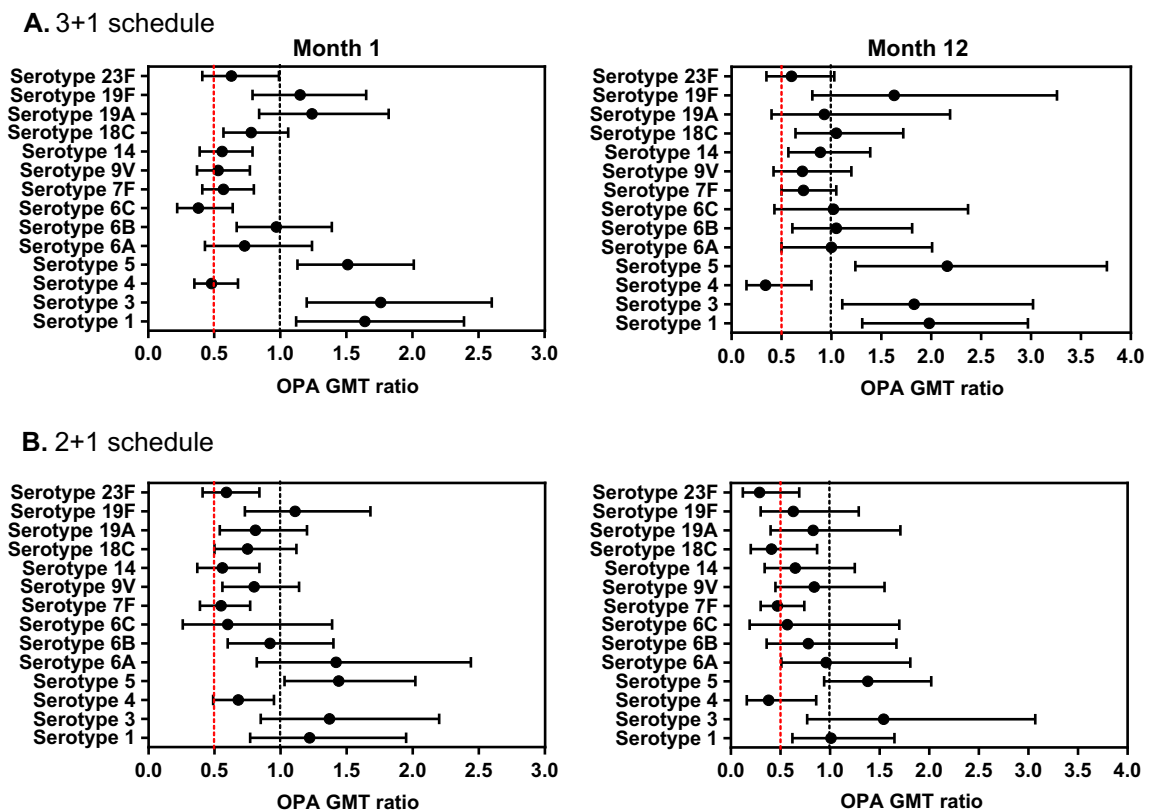
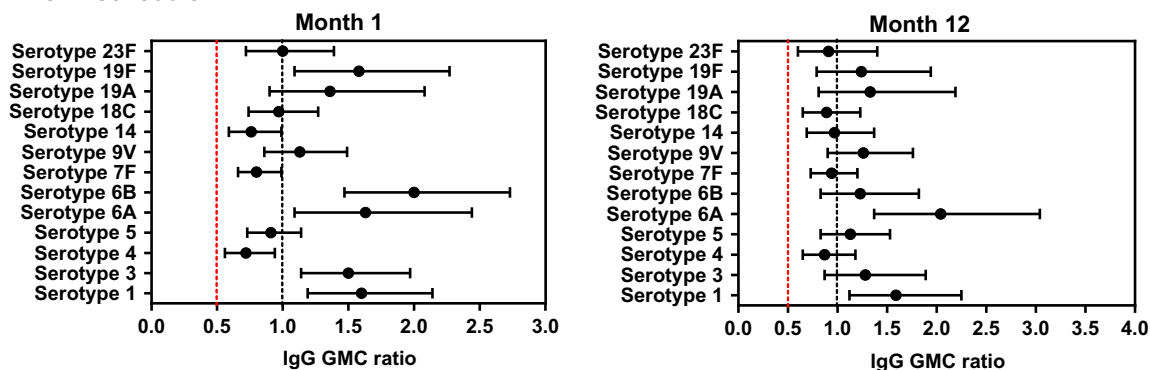


Fig. 2. PHiD-CV/PCV13 ratio of anti-pneumococcal OPA titers 1 and 12 months after the PCV13 booster dose. Ratios of OPA geometric mean titer in PHiD-CV over PCV13 recipients at 1 month and 12 months after the booster dose (with the respective baseline value as covariate). The error bars indicate 95% confidence intervals, the red dotted line the arbitrary margin for non-inferiority, and the black dotted line a ratio of 1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

A. 3+1 schedule



B. 2+1 schedule

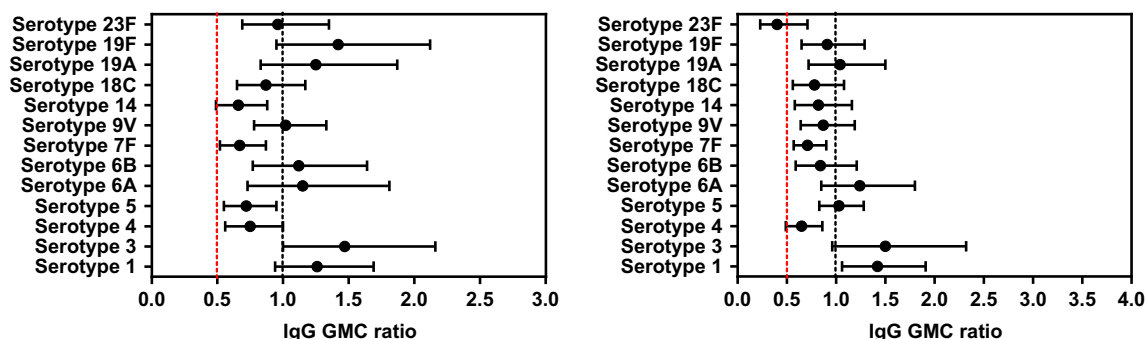


Fig. 3. PHiD-CV/PCV13 ratio of anti-pneumococcal IgG concentrations 1 and 12 months after the PCV13 booster dose. Ratios of IgG geometric mean concentration in PHiD-CV over PCV13 recipients at 1 month and 12 months after the booster dose (with the respective baseline value as covariate). The error bars indicate 95% confidence intervals, the red dotted line the arbitrary margin for non-inferiority, and the black dotted line a ratio of 1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

were non-inferior to those in PCV13-primed children irrespective of schedule. Non-inferiority was also suggested at 12 months.

3.3. OPA responses to the other vaccine serotypes

One month post-booster, the OPA titers increased substantially for all the other serotypes in both studies ([Supplementary Table 1](#)). With both schedules, OPA GMTs were similar between groups for serotypes 1, 3, 5, 6B, 18C, 19F, and 23F but, with the 3 + 1 schedule, they were higher in the PCV13 group for serotypes 4, 6A, 6C, 7F, 9V, and 14. In both studies, all subjects had OPA titers ≥ 8 in the two groups for all serotypes, except for serotypes 3 and 6A ($\geq 97.4\%$) ([Supplementary Table 2](#)). For the vaccine-related serotype 6C, the proportions were slightly lower in the 2 + 1 schedule than in the 3 + 1 schedule (92.1–93.9% vs. 98–100%).

With the 3 + 1 schedule, OPA GMTs in PHiD-CV-primed subjects were similar to those in PCV13-primed subjects for six serotypes (1, 3, 5, 6B, 18C, and 19F) but were inferior for seven (4, 6A, 6C, 7F, 9V, 14, 23F) ([Fig. 2](#)). With the 2 + 1 schedule, OPA GMTs were similar in PHiD-CV-primed subjects for seven serotypes (1, 3, 5, 6A, 6B, 9V, and 19F) and inferior for six (4, 6C, 7F, 14, 18C, and 23F).

3.4. Anti-pneumococcal IgG responses to the other serotypes

IgG GMCs increased for all serotypes at 1 month post-booster in both studies ([Supplementary Table 3](#)). GMCs were comparable between groups for all vaccine serotypes, except for serotypes 6B and 19F, which were higher in the PHiD-CV group than in the

PCV13 group irrespective of priming schedule. All subjects had antibody GMCs ≥ 0.35 mg/mL for serotypes 1, 4, 7F, 14, 18C, and 19F irrespective of priming schedule and vaccine ([Supplementary Table 4](#)). For serotypes 3, 5, 6A, 6B, 9V, and 23F, the percentages ranged between 84.2% and 100.0%. Post-PCV13 booster responses were non-inferior for all serotypes irrespective of priming schedule and vaccine, except for serotype 14 in the PHiD-CV-primed 2 + 1 schedule ([Fig. 3](#)).

3.5. Persistence of the OPA and IgG responses to the other vaccine serotypes

Twelve months after the booster dose, the OPA GMTs and the IgG GMCs decreased approximately to baseline levels for all serotypes following either schedule ([Supplementary Tables 1 and 3](#)). Although the proportions of subjects with OPA GMTs ≥ 8 also decreased over time, they remained $>90\%$ for serotypes 6B, 7F, and 23F; $>80\%$ for serotypes 6A, 9V, 18C, 19A, and 19F; and between 47% and 94% for serotypes 4, 5, 6C, and 14 ([Supplementary Table 2](#)). The lowest proportions of subjects with GMTs ≥ 8 were for serotypes 1 and 3 (13–61%). Overall, there were no major differences between children primed with different vaccines, except for serotype 1 in the 3 + 1 schedule (43.1% in the PHiD-CV group vs. 13.3% in the PCV13 group) and serotype 4 in the 2 + 1 schedule (47.4% in the PHiD-CV group vs. 81.6% in the PCV13 group). The proportions of subjects with IgG concentrations above the $0.35 \mu\text{g/mL}$ threshold also decreased over time and were $\geq 76.5\%$ for serotypes 6A, 6B, 7F, 14, and 19F and between 15.7% and 65.8% for serotypes 1, 3, 4, 5, 9V, 18C, and 23F at 12 months

Table 2
Safety and reactogenicity by subject.

Symptom	3 + 1 schedule (Czech Republic)						2 + 1 schedule (Slovakia)					
	PHiD-CV (N = 53)		PCV13 (N = 45)		Total (N = 98)		PHiD-CV (N = 39)		PCV13 (N = 50)		Total (N = 89)	
	n subjects (%)	n AEs	n subjects (%)	n AEs	n subjects (%)	n AEs	n subjects (%)	n AEs	n subjects (%)	n AEs	n subjects (%)	n AEs
Reactogenicity												
<i>Local reactions</i>												
Any	45 (84.9%)	–	36 (80.0%)	–	81 (82.7%)	–	32 (82.1%)	–	45 (90.0%)	–	77 (86.5%)	–
Grade 3	1 (1.9%)	1	0	0	1 (1.0%)	1	2 (5.1%)	2	1 (2.0%)	1	3 (3.4%)	3
<i>Systemic reactions</i>												
Any	44 (83.0%)	–	37 (82.2%)	–	81 (82.7%)	–	29 (74.4%)	–	45 (90.0%)	–	74 (83.1%)	–
Grade 3	0	0	2 (4.4%)	2	2 (2.0%)	2	1 (2.6%)	2	1 (2.0%)	1	2 (2.2%)	3
Safety												
<i>Adverse events</i>												
Any	14 (26.4%)	23	26 (57.8%)	60	40 (40.8%)	83	15 (38.5%)	32	21 (42.0%)	31	36 (40.4%)	63
Severe	1 (1.9%)	1	3 (6.7%)	4	4 (4.1%)	5	0	0	1 (2.0%)	1	1 (1.1%)	1
Vaccination-related	0	0	2 (4.4%)	2	2 (2.0%)	2	1 (2.6%)	2	1 (2.0%)	1	2 (2.2%)	3
<i>Serious adverse events</i>												
Any	1 (1.9%)	1	3 (6.7%)	3	4 (4.1%)	4	2 (5.1%)	4	1 (2.0%)	1	3 (3.4%)	5

AEs, adverse events; PCV13, 13-valent pneumococcal conjugate vaccine; PHiD-CV, 10-valent pneumococcal non-typeable *Haemophilus influenzae* protein D conjugate vaccine.

post-booster (Supplementary Table 4). In both studies, there were no major differences between vaccine groups.

Twelve months post-booster, the OPA GMTs in PHiD-CV-primed subjects were similar to those in PCV13-primed subjects for serotypes 1, 3, 5, 6B, 14, 18C, and 19F with the 3 + 1 schedule and for serotypes 1, 3, 5, and 6A with the 2 + 1 schedule (Fig. 2). The IgG concentrations after PHiD-CV priming were similar for all serotypes with the 3 + 1 schedule and for all serotypes except serotypes 4 and 23F with the 2 + 1 schedule (Fig. 3).

3.6. Safety and reactogenicity of the PCV13 booster dose

Overall, solicited local and systemic reactions were reported in 82.7% of subjects in the 3 + 1 schedule and 86.5% in the 2 + 1 schedule (Table 2). In the 3 + 1 schedule, one subject had a grade 3 local reaction (pain) and two subjects had grade 3 systemic reactions (irritability/fussiness in both). In the 2 + 1 schedule, three subjects had grade 3 pain at the injection site. One subject had one grade 3 systemic reaction (irritability/fussiness) and one subject had two (irritability/fussiness and loss of appetite).

Overall, approximately 40% of subjects in both studies had unsolicited AEs within 1 month after the PCV13 booster dose (Table 2). These AEs were mainly respiratory conditions and of mild intensity. In the 3 + 1 schedule, the frequency of subjects reporting AEs and the number of AEs were approximately two fold lower in the PHiD-CV group than in the PCV13 group. Two AEs in the PCV13 group, erythema and rhinitis, were considered vaccine-related by the investigators. They began within 3 days of the booster dose, were of mild intensity, and resolved without sequelae. In the 2 + 1 schedule, one subject in the PHiD-CV group had two AEs (acute gastroenteritis and dehydration) and one subject in the PCV13 group had one AE (sideropenic anemia) considered vaccine-related.

Four subjects in the 3 + 1 schedule had SAEs (acute laryngitis, head and neck burn, herniotomy in the PCV13 group; falling on head in the PHiD-CV group) and three in the 2 + 1 schedule (severe dyspepsia in the PCV13 group; respiratory tract inflammation [rhinopharyngitis, obstructive bronchitis, and unilateral pneumonia] and acute gastroenteritis with dehydration in the PHiD-CV group). None of the SAEs were considered related to the study vaccine.

4. Discussion

Our results show that a PCV13 booster dose is immunogenic and well tolerated in PCV13- or PHiD-CV-primed children irrespective of the priming schedule. The primary and secondary confirmatory objectives were met for both studies. One month post-PCV13 booster, the OPA and IgG immune responses to serotype 19A in PHiD-CV-primed children were not inferior to those in PCV13-primed children, as part of either 3 + 1 or 2 + 1 schedules.

An IgG concentration of 0.35 µg/mL correlates with protection at a population level and is therefore used as a non-inferiority threshold for evaluating PCVs [8]. However, because antibody levels measured by ELISA do not necessarily reflect functional IgG, the ability to kill pneumococci as measured by in vitro OPA assay is considered important in evaluating PCVs [24,25]. Whereas IgG responses after PCV13 revealed some serotype-specific differences dependent on the priming vaccine, nearly all subjects had OPA titers ≥ 8 , suggesting that quantitatively lower responses were still functional. The IgG GMCs and the OPA GMTs decreased over time but the proportions of subjects with immune responses above the thresholds remained high for most serotypes at 12 months post-booster. This suggests that, although the immune responses waned after the booster dose, this decline may be of limited clinical relevance for most serotypes. Further studies of long-term immune response persistence and epidemiological studies are thus needed to determine if PCV13 vaccination correlates with long-term protection against *S. pneumoniae*.

Although serotypes 3, 6A, and 19A are not included in PHiD-CV, high IgG GMCs and OPA GMTs against these serotypes were found after the PCV13 booster dose in PHiD-CV-primed subjects. Only for serotype 6A and only in children primed with 3 doses of PCV13 were OPA responses to the PCV13 booster superior to those seen in PHiD-CV-primed children. Mixed PHiD-CV/PCV13 schedules may therefore offer a reasonable alternative, including in settings with high occurrence of 19A disease. In Quebec, Canada, a mixed schedule consisting of 2 primary PHiD-CV doses followed by a PCV13 booster provided similar protection against overall and vaccine-type invasive pneumococcal disease, including for serotype 19A, as compared to schedules using only PHiD-CV or PCV13 [26].

One previous study evaluated the interchangeability of PHiD-CV and PCV13 when given as booster doses in mixed schedules

[16]. The study showed that a PHiD-CV booster is generally less immunogenic than a PCV13 booster in PCV13-primed children. Our studies evaluated for the first time a booster dose with PCV13 in children primed with PHiD-CV or PCV13. Our results suggested that PCV13 induces strong immune responses against vaccine serotypes irrespective of the priming vaccine and can be used as a booster in the context of mixed schedules.

The reactogenicity and safety profiles of the PCV13 booster dose were comparable for both schedules. In both studies, the frequency and severity of the solicited reactions and the unsolicited AEs in both study groups corresponded to those commonly observed after administration of vaccines. These results suggest that using a PCV13 booster after PHiD-CV priming does not raise any safety issues. Indeed, a switch from PHiD-CV to PCV13 has been introduced in the immunization programs in several countries [27,28], and no safety issues were raised during the transition period.

Our studies have some limitations. The calculated target sample size of 42 evaluable children per group was not reached in the PHiD-CV group in Slovakia (39 children) because enrolment was very slow in this group due to hesitancy towards mixed schedule and preference towards a full schedule with PCV13. In addition, the sample size, which was calculated based on serotype 19A responses, may not have been sufficient for testing non-inferiority of other serotypes. Furthermore, multiplicity of comparisons was not taken into account, which could influence the results in either direction. The 3 + 1 and 2 + 1 schedules were assessed in two different studies conducted in different countries, which may also limit comparisons between the studies. However, the immunological analyses were performed by the same laboratory to limit the risk of bias. Finally, the different coadministration of DTaP-HBV-IPV between the two countries may have influenced the immunogenicity of the PCV13 booster, although immune interference between these vaccines has not been observed [29]. A strength of our studies is that the evaluated schedules are not strictly timed, suggesting that our results are applicable to real-life vaccination practice.

In conclusion, our results suggest that a PCV13 booster in children primed with 2 or 3 doses of PHiD-CV could be a promising vaccination strategy to improve clinical protection against three additional serotypes, although the clinical relevance of the observed differences in IgG and OPA responses between groups is currently unknown.

Acknowledgements

The authors thank the parents, children, and local investigators in the Czech Republic: Věra HVÍŽDALOVÁ (Pardubice), Renáta RŮŽKOVÁ and Soňa KNĚBLOVÁ (Praha), Daniel DRAŽAN and Ludmila PLOCKOVÁ (Jindřichův Hradec), Pavla DRNKOVÁ and Marcela DANIELOVÁ (Ústí n. Labem), Kateřina ŠTICHHAUEROVÁ (Pardubice), Věra RYVOLOVÁ (Havlíčkův Brod), Zdeněk SLAVÍK (Hradec Králové); and in the Slovak Republic: Marta ŠPÁNIKOVÁ, Elena PROKOPOVÁ, Zuzana KOŠTÁLOVÁ, and Beáta KARTOUSOVÁ (Bratislava), Helena ŠKERLÍKOVÁ and Mária ŽÁNOVÁ (Dolný Kubín), Jana SNOPKOVÁ and Beáta BALLUCHOVÁ (Košice).

Statistical analysis was provided by Pharmnet (the Czech Republic). Medical writing assistance was provided by Dr. Julie Harriague (4Clinics, France).

Authors contributions

I.U. and R.P. were the lead investigators and contributed to protocol design, data collection, data analysis, and data interpretation. K. P. and P.K. contributed to protocol design, data analysis, and data interpretation. D.G. and L.R. contributed to protocol design, data collection, and interpretation. All authors contributed to the writ-

ing of the manuscript, provided critical comments on manuscripts drafts, and approved the final version of the manuscript and the decision to submit for publication.

Conflicts of interest

R.P. reports grants from the GSK group of companies during the conduct of the study and grants from the GSK group of companies, Novartis, and Sanofi Pasteur outside the submitted work. I.U. has received grants and lecture fees from GSK, Pfizer, MSD, Novartis, and Sanofi Pasteur outside the submitted work. L.R., K.P., and P.K. have nothing to disclose. The University College London laboratory conducts contract research for PCV vaccine manufacturers including the GSK group of companies. D.G. has participated in Advisory Boards for vaccine manufacturers including the GSK group of companies.

Source of funding

This work was supported by Biovomed, which paid for all costs associated with the development and the publishing of the present manuscript. The sponsor was involved in all stages of the study conduct and analysis. The studies were performed with financial support from GlaxoSmithKline Biologicals S.A. D.G. is partially funded by the UK National Institute for Health Research.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.vaccine.2017.07.103>.

References

- [1] Conklin L, Loo JD, Kirk J, Fleming-Dutra KE, Deloria Knoll M, Park DE, et al. Systematic review of the effect of pneumococcal conjugate vaccine dosing schedules on vaccine-type invasive pneumococcal disease among young children. *Pediatr Infect Dis J* 2014;33(Suppl 2):S109–18. <http://dx.doi.org/10.1097/inf.0000000000000073>.
- [2] Fitzwater SP, Chandran A, Santosham M, Johnson HL. The worldwide impact of the seven-valent pneumococcal conjugate vaccine. *Pediatr Infect Dis J* 2012;31(5):501–8. <http://dx.doi.org/10.1097/INF.0b013e31824de9f6>.
- [3] Hausdorff WP. Invasive pneumococcal disease in children: geographic and temporal variations in incidence and serotype distribution. *Eur J Pediatr* 2002;161(Suppl 2):S135–9. <http://dx.doi.org/10.1007/s00431-002-1066-x>.
- [4] Moore MR, Gertz Jr RE, Woodbury RL, Barkocy-Gallagher GA, Schaffner W, Lexau C, et al. Population snapshot of emergent *Streptococcus pneumoniae* serotype 19A in the United States, 2005. *J Infect Dis* 2008;197(7):1016–27. <http://dx.doi.org/10.1086/528996>.
- [5] Centers for Disease Control and Prevention. Direct and indirect effects of routine vaccination of children with 7-valent pneumococcal conjugate vaccine on incidence of invasive pneumococcal disease—United States, 1998–2003. *MMWR Morb Mortal Wkly Rep* 2005;54(36):893–7.
- [6] Weil-Olivier C, van der Linden M, de Schutter I, Dagan R, Mantovani L. Prevention of pneumococcal diseases in the post-seven valent vaccine era: a European perspective. *BMC Infect Dis* 2012;12:207. <http://dx.doi.org/10.1186/1471-2334-12-207>.
- [7] Reinert R, Jacobs MR, Kaplan SL. Pneumococcal disease caused by serotype 19A: review of the literature and implications for future vaccine development. *Vaccine* 2010;28(26):4249–59. <http://dx.doi.org/10.1016/j.vaccine.2010.04.020>.
- [8] World Health Organization. Recommendations for the production and control of pneumococcal conjugate vaccines. WHO Technical Report Series (927, Annex 2). Switzerland: Geneva; 2005.
- [9] World Health Organization. Pneumococcal vaccines WHO position paper—2012. *Wkly Epidemiol Rec* 2012;87(14):129–44.
- [10] Spijkerman J, Veenhoven RH, Wijnga-Monsuur AJ, Elberse KE, van Gageldonk PG, Knol MJ, et al. Immunogenicity of 13-valent pneumococcal conjugate vaccine administered according to 4 different primary immunization schedules in infants: a randomized clinical trial. *JAMA* 2013;310(9):930–7. <http://dx.doi.org/10.1001/jama.2013.228052>.
- [11] Deloria Knoll M, Park DE, Johnson TS, Chandir S, Nonyane BA, Conklin L, et al. Systematic review of the effect of pneumococcal conjugate vaccine dosing schedules on immunogenicity. *Pediatr Infect Dis J* 2014;33(Suppl 2):S119–29. <http://dx.doi.org/10.1097/inf.0000000000000079>.

- [12] Goldblatt D, Southern J, Ashton L, Richmond P, Burbidge P, Tasevska J, et al. Immunogenicity and boosting after a reduced number of doses of a pneumococcal conjugate vaccine in infants and toddlers. *Pediatr Infect Dis J* 2006;25(4):312–9. <http://dx.doi.org/10.1097/01.inf.0000207483.60267.e7>.
- [13] Whitney CG, Goldblatt D, O'Brien KL. Dosing schedules for pneumococcal conjugate vaccine: considerations for policy makers. *Pediatr Infect Dis J* 2014;33(Suppl 2):S172–81. <http://dx.doi.org/10.1097/inf.0000000000000076>.
- [14] Prymula R, Schuerman L. 10-valent pneumococcal nontypeable *Haemophilus influenzae* PD conjugate vaccine: Synflorix. *Expert Rev Vaccines* 2009;8(11):1479–500. <http://dx.doi.org/10.1586/erv.09.113>.
- [15] Paradiso PR. Advances in pneumococcal disease prevention: 13-valent pneumococcal conjugate vaccine for infants and children. *Clin Infect Dis* 2011;52(10):1241–7. <http://dx.doi.org/10.1093/cid/cir142>.
- [16] Trück J, Jawad S, Goldblatt D, Roalfe L, Snape MD, Voysey M, et al. The antibody response following a booster with either a 10- or 13-valent pneumococcal conjugate vaccine in toddlers primed with a 13-valent pneumococcal conjugate vaccine in early infancy. *Pediatr Infect Dis J* 2016;35(7):787–93. <http://dx.doi.org/10.1097/INF.0000000000001180>.
- [17] Wijmenga-Monsuur AJ, van Westen E, Knol MJ, Jongerius RM, Zancolli M, Goldblatt D, et al. direct comparison of immunogenicity induced by 10- or 13-valent pneumococcal conjugate vaccine around the 11-month booster in Dutch infants. *PLoS One* 2015;10(12):e0144739. <http://dx.doi.org/10.1371/journal.pone.0144739>.
- [18] van Westen E, Wijmenga-Monsuur AJ, van Dijken HH, van Gaans-van den Brink JA, Kuipers B, Knol MJ, et al. Differential B-cell memory around the 11-month booster in children vaccinated with a 10- or 13-valent pneumococcal conjugate vaccine. *Clin Infect Dis* 2015;61(3):342–9. <http://dx.doi.org/10.1093/cid/civ274>.
- [19] Wernette CM, Frasch CE, Madore D, Carlone G, Goldblatt D, Plikaytis B, et al. Enzyme-linked immunosorbent assay for quantitation of human antibodies to pneumococcal polysaccharides. *Clin Diagn Lab Immunol* 2003;10(4):514–9. <http://dx.doi.org/10.1128/CDLI.10.4.514-519.2003>.
- [20] Rose CE, Romero-Steiner S, Burton RL, Carlone GM, Goldblatt D, Nahm MH, et al. Multilaboratory comparison of *Streptococcus pneumoniae* opsonophagocytic killing assays and their level of agreement for the determination of functional antibody activity in human reference sera. *Clin Vaccine Immunol* 2011;18(1):135–42. <http://dx.doi.org/10.1128/CVI.00370-10>.
- [21] Grant LR, O'Brien SE, Burbidge P, Haston M, Zancolli M, Cowell L, et al. Comparative immunogenicity of 7 and 13-valent pneumococcal conjugate vaccines and the development of functional antibodies to cross-reactive serotypes. *PLoS One* 2013;8(9):e74906. <http://dx.doi.org/10.1371/journal.pone.0074906>.
- [22] Andrews NJ, Waight PA, Burbidge P, Pearce E, Roalfe L, Zancolli M, et al. Serotype-specific effectiveness and correlates of protection for the 13-valent pneumococcal conjugate vaccine: a postlicensure indirect cohort study. *Lancet Infect Dis* 2014;14(9):839–46. [http://dx.doi.org/10.1016/S1473-3099\(14\)70822-9](http://dx.doi.org/10.1016/S1473-3099(14)70822-9).
- [23] Schuerman L, Wysocki J, Tejedor JC, Knuf M, Kim KH, Poolman J. Prediction of pneumococcal conjugate vaccine effectiveness against invasive pneumococcal disease using opsonophagocytic activity and antibody concentrations determined by enzyme-linked immunosorbent assay with 22F adsorption. *Clin Vaccine Immunol* 2011;18(12):2161–7. <http://dx.doi.org/10.1128/CVI.05313-11>.
- [24] Johnson SE, Rubin L, Romero-Steiner S, Dykes JK, Pais LB, Rizvi A, et al. Correlation of opsonophagocytosis and passive protection assays using human anticapsular antibodies in an infant mouse model of bacteremia for *Streptococcus pneumoniae*. *J Infect Dis* 1999;180(1):133–40. <http://dx.doi.org/10.1086/314845>.
- [25] Romero-Steiner S, Frasch CE, Carlone G, Fleck RA, Goldblatt D, Nahm MH. Use of opsonophagocytosis for serological evaluation of pneumococcal vaccines. *Clin Vaccine Immunol* 2006;13(2):165–9. <http://dx.doi.org/10.1128/CVI.13.2.165-169.2006>.
- [26] Deceuninck G, De Serres G, Boulianne N, Lefebvre B, De Wals P. Effectiveness of three pneumococcal conjugate vaccines to prevent invasive pneumococcal disease in Quebec, Canada. *Vaccine* 2015;33(23):2684–9. <http://dx.doi.org/10.1016/j.vaccine.2015.04.005>.
- [27] Institut national de santé publique du Québec. Comité sur l'immunisation du Québec. Évaluation de deux nouveaux vaccins pneumococciques conjugués pour l'immunisation des enfants au Québec; 2010. Available from: https://www.inspq.qc.ca/pdf/publications/1172_Eval2VaccinsPneumocoConjuguésImmuEnfants.pdf. Accessed: 01 Feb 2017.
- [28] Leach AJ, Wigger C, Beissbarth J, Woltring D, Andrews R, Chatfield MD, et al. General health, otitis media, nasopharyngeal carriage and middle ear microbiology in Northern Territory aboriginal children vaccinated during consecutive periods of 10-valent or 13-valent pneumococcal conjugate vaccines. *Int J Pediatr Otorhinolaryngol* 2016;86:224–32. <http://dx.doi.org/10.1016/j.ijporl.2016.05.011>.
- [29] Nunes MC, Madhi SA. Review on the immunogenicity and safety of PCV-13 in infants and toddlers. *Expert Rev Vaccines* 2011;10(7):951–80. <http://dx.doi.org/10.1586/erv.11.76>.